

BAPPADITYA KUILYA

Kolkata, India | +91 9883867522 | bappadityakuilya@gmail.com
[linkedin.com/in/bappaditya-kuiilya](https://www.linkedin.com/in/bappaditya-kuiilya) | github.com/Bappaditya-kuiilya | bappadityakuilya.is-a.dev

SUMMARY

Computer Science undergraduate (CGPA 8.11) building production backend systems and agentic AI applications. Shipped a security-audited multi-agent orchestration engine (124 tests, 9.5/10 security score) plus deployed full-stack products with real users. Strong foundation in DSA, OOP, and system design.

TECHNICAL SKILLS

- **Languages:** Java, Python, JavaScript, C, SQL
- **Backend & AI:** REST APIs, LLM APIs (Groq, Cohere, Gemini), Prompt Engineering, Multi-Agent Systems, TensorFlow, Neural Networks
- **Tools & Platforms:** Git, Docker, React, Node.js, Streamlit, WebRTC, Linux/WSL, Vercel, Render
- **Core CS:** Data Structures & Algorithms, Object-Oriented Programming, System Design, DBMS, Operating Systems

TECHNICAL PROJECTS

Multi-Tool Orchestrator Agent | [github.com/Bappaditya-kuiilya/Multi-tool Orchestrator Agent](https://github.com/Bappaditya-kuiilya/Multi-tool_Orchestrator_Agent) | *Python, Pydantic, GitHub Actions*

- Built a permission-scoped multi-tool orchestration engine (~1,500 lines) with a YAML-based tool registry, exact + semantic (TF-IDF) capability routing, and scoped permission tokens
- Hardened the execution layer against scope escalation, token inflation, and CPU DoS attacks; shipped with a documented 9.5/10 security audit and 124 automated tests passing at 0 warnings
- Implemented sequential and parallel executors, an in-process pub/sub message queue, and JSONL audit trails, wired into a CI pipeline (GitHub Actions: pytest + compileall)

RegWatch — Agentic Regulatory Intelligence Platform | github.com/Bappaditya-kuiilya/RegWatch | *Python, Multi-Agent Pipeline, Streamlit, Docker*

- Built a multi-agent system that autonomously ingests and reasons over regulatory changes to deliver compliance intelligence for Indian MSMEs
- Designed an ingestion → graph → core reasoning agent pipeline integrating multiple LLM providers (Groq, Cohere, Gemini) with a scheduled 24-hour autonomous refresh cycle
- Containerized with Docker and deployed on Streamlit Community Cloud with separate demo and live-scrape modes for reliable, repeatable runs

Peek — Encrypted Peer-to-Peer File Transfer | github.com/Bappaditya-kuiilya/Peek | *React, Node.js, WebRTC, Web Crypto API*

- Built a browser-based, installation-free file transfer tool with client-side AES-256-GCM end-to-end encryption; encryption keys never reach the server
- Implemented WebRTC peer-to-peer transfer with automatic fallback to an encrypted relay, plus two-way sessions and clipboard sync
- Deployed a full production stack (Vercel frontend, Render relay) with zero data persistence and a 60-minute session TTL

EDUCATION

Bachelor of Technology in Computer Science Engineering — CGPA: 8.11/10

University of Engineering & Management, Kolkata | Data Structures, Algorithms, OOP, DBMS, Operating Systems

ACHIEVEMENTS

- **Hackathon Winner (1x):** Won 1st place, demonstrating rapid prototyping and team collaboration under time constraints
- **Problem Solving:** Solved 160+ LeetCode problems across DSA, dynamic programming, and optimization while maintaining an 8.11 CGPA
- **Active Learner:** Regular participant in tech bootcamps, hackathons, and quizzing; strong interest in AI systems and scalable products